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Long-term evaluation of a health education programme for elderly persons with visual impairment. A randomized study

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Abstract

In order to implement evidence-based practice, a randomized study was set up to evaluate the ADL- based Health Education Programme 'Discovering new ways' for elderly persons with age-related macular degeneration.
Purpose: To investigate the impact of this program on perceived security in the performance of daily activities 28 months after the intervention.
Method: Two-hundred and twenty-nine persons randomized to either the Health Education programme or an Individual Intervention Programme participated in the study. At the 28-month follow-up there was a dropout of 98 persons and the results are based on 62 persons participating in the Health Education Programme and 69 persons in the Individual Intervention Programme.
Results: There were statistically significant differences in perceived security between the groups in 15 out of 28 daily activities. Furthermore, the Health Education Group showed a significant tendency towards an improved level of security while the Individual Intervention Group tended to deteriorate.
Conclusions: The findings provide strong support for the long-term effect of the programme and for the implementation of evidence-based practice. The study corroborates the effectiveness of the Health Education Programme in enhancing security and hindering a progressive decline in perceived security in daily activities.

Introduction

A profession has a responsibility to ensure the best practice that has proved to be effective and to use evidence in practice.[1, 2] In order to implement evidence-based practice, a randomized study was set up to evaluate the ADL- based Health Education Programme 'Discovering new ways'.[3] The main goal of the programme, which targets elderly persons with age-related macular degeneration (AMD), is to sustain and restore the participants performance of their daily activities and to enable the early detection of those who perceive insecurity when performing daily activities. The programme was evaluated at a 4-month follow-up in terms of perceived security in daily activities[4] and was found to enhance security or hinder a progressive decline in perceived security in daily activities. Further research is needed to investigate whether the changes will continue in the long term. The aim of the present study is thus to examine the long-term stability of perceived security in daily activities.

AMD is a severe, progressive, chronic eye disease of the central part of the retina (macula), which rarely leads to total blindness[5] and is the most common cause of irreversible visual impairment among elderly persons past the age of 60. Studies show that about 30% of those who are 75 years or older are affected by AMD.[6, 7] Elderly persons with AMD[3] claim to have problems in several activities as a result of their visual impairment. A population study[8] of 85-year-olds with AMD found this group to be more dependent on personal help in both personal and instrumental daily activities compared with persons of the same age with normal vision. Since no cure exists for the majority of people with AMD, this chronic impairment thus represents an important public health problem.

The development and evaluation of the Health Education Programme 'Discovering new ways' occurred in three stages, pre-intervention stage, intervention and

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post-intervention stage. The purpose of the pre-intervention stage was to estimate the need for intervention in the target group. A population study[8] was conducted to determine the prevalence of problems in daily activities. Focus group methodology[9] was applied at this stage to learn how the target group described and perceived their disease, including how the disease had changed their daily activities. The intervention stage came next and was designed to further develop and improve the programme. Focus group methodology was used as a tool for process evaluation and the focus was on the participant's reaction to the programme itself and on finding out why the programme succeeded or did not succeed. The purpose of the last stage was to evaluate the programme. Focus group methodology was used to develop an evaluative instrument. The items and the scale were selected on the basis of the focus group study in the pre-intervention stage[10] where elderly persons with AMD describe a feeling of insecurity performing daily activities. The instrument was used to evaluate the effectiveness of the programme at a four- month follow-up.[4] Measuring the effectiveness of the intervention will continue to offer controlled evidence of its benefits.

The purpose of the present study was to investigate the impact of the Health Education Programme on perceived security in the performance of daily activities by a randomized study of two groups of elderly 28 months after two types of intervention. The differences in perceived security between the groups after 28 months as well as changes over time within each group will be reported.

Methods

STUDY DESIGN

The study was carried out at a low vision clinic at a university-affiliated hospital in Sweden as a randomized longitudinal study. The study was designed so that the follow-ups took place 1, 4, 16 and 28 months after the intervention was completed. The Ethics Committee of the faculty of medicine at the university approved the study. All persons gave their informed consent to taking part in the study.

STUDY POPULATION

The target group comprized all persons referred by an ophthalmologist to rehabilitation and who were attending the low vision clinic for the first time. The inclusion criteria were as follows: 65 years or older, living at home, aged-related macular degeneration as the primary diagnosis, a distance visual acuity of the better eye with best correction no lower than 0.1 and ability to participate in group discussions. A total of 312 persons referred to the low vision clinic between January 1996 and December 1997 met the study criteria and were consecutively asked to participate in the study. Of these, 229 (73%) persons actually participated. Among the non-participants, the main reason given for not participating in the study was poor health (59%).

There was a dropout of 98 persons (Individual Intervention Programme $n = 51$, Health Education Programme $n = 47$) at the 28-month follow – up. The present study is based on those 131 persons (Individual Intervention Programme $n = 69$, Health Education Programme $n = 62$) who were still participating at that time.

STUDY PROCEDURE

The participants were randomly assigned, according to a random number table, either to the Health Education Programme or to an Individual Intervention Programme that was standard at the low vision clinic. The occupational therapists that collected the data were not blinded to the composition of the groups but were not involved in the program.

Assessments at baseline and at the 28-month follow-up were both made when the participants in the Health Education Programme and in the Individual Intervention Programme attended the low vision clinic. The study procedure did not differ between the programmes. The independently registered occupational therapists interviewed the participants according to a structured protocol, consisting of questions about marital status, living arrangements, social service, granted public transportation and perceived health. An assessment was also made of perceived security in performing daily activities, further described below.[10] An optometrist made the optical assessment at the visit. Visual acuity was tested with a letter chart (Monoyer-Granström, Kifa), graded 0.1 – 1.0 at a distance of 5 m, with the person's own glasses and with best refraction.

OUTCOME ASSESSMENTS

The primary outcome measure was perceived security in performing daily activities.[3, 10] The instrument has been tested for reliability and responsiveness and was found to have a high level of within-subject stability and responsiveness. The instrument has also face and content validity.[10]

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The instrument is a questionnaire, which consists of seven areas with a total of 28 items. The areas are as follows: meals, self-care and care of clothing, communication, cleaning, mobility, shopping and financial management. The perceived security in performing each specified task is rated by means of a four-point ordinal scale: very insecure, insecure, quite secure, secure. The instrument is self-rated, and the participants were instructed to fill in the questionnaire on their own after instructions from the occupational therapists. For those persons who had trouble reading, the rater read each question and each alternative response. The participants were also given the response alternatives in enlarged print. The participants were instructed to assess their actual performance and how they managed at that very moment. They had to perform the activity independently, and in those activities where they asked for help in a limited part of the performance were regarded as independent. They were instructed to leave out any activities that they did not perform.

BASELINE CHARACTERISTICS

The non-participants ($n = 83$) differed statistically significantly from the participants ($n = 131$), with a higher proportion of persons who had been granted public transportation service, as shown in table 1. In the other characteristics; gender, age, living alone, social service, perceived good health and visual acuity there were no statistical significant differences. There were no statistical significant difference between the dropouts ($n = 98$) and the participants.

There were no statistical significant differences at baseline between the Health Education Group and the Individual Intervention Group concerning; gender, age, social service, public transportation service, perceived good health and visual acuity with the exception of persons living alone. The Health Education Group had a significant higher proportion of persons living alone (69%; 52%, confidence interval 95%: 1–33).

INTERVENTION WITH THE HEALTH EDUCATION PROGRAMME: DISCOVERING NEW WAYS

The programme was carried out with groups of four to six persons with age-related macular degeneration. Twenty groups were formed consecutively during the study period. The intervention period was 8 weeks long, and the groups met for 2 h once a week. The programme and its theoretical basis have been described elsewhere.[3, 11] The groups were led by experienced occupational therapists trained in leading the groups. The occupational therapist provided information and skills training and guided and encouraged the group participants in the learning process. Other health professionals such as an ophthalmologist, optometrist, low vision therapist and lighting expert were invited to give information.

The information and the skills training given by the occupational therapist proceeded from different strategies in daily activities described by elderly persons with age-related macular degeneration.[3] The strategies were used as a problem-solving model, and the participants were enabled to use the model as a way of thinking when performing daily activities.

A booklet[12] containing the information given by health professionals as well as the information about daily activities was produced and used in the Health Education Programme. The participants were asked to

Table 1 Baseline characteristics of participants, non-participants and drop-outs. Confidence interval (CI) for proportions and their differences and difference between means are indicated at 95% between participants and; non-participants and drop-outs respectively. Significant differences are indicated in bold.

	Participants n = *131*	*Non-participants* n = *83*	*95% CI*	*Drop-outs* n = *98*	*95% CI*
Females, %	74	71	− 9 to 15	62	0 to 24
Age, mean	78	82	− 4 to 12	81	− 4 to 10
(range)	(66–91)	(69–96)		(68–94)	
Living alone, %	60	58	− 12 to 16	64	− 9 to 17
Public transportation service, %	37	54	**3 to 31**	49	− 1 to 25
Social service, %	18	25	− 4 to 18	21	− 7 to 13
Perceived good health, %	86	78	− 3 to 19	79	− 3 to 17
Visual acuity, median,	0.3	0.3		0.3	
(range)	(1.0–0.1)	(0.8–0.1)		(1.0–0.1)	

prepare themselves before participating each session by reading the relevant chapter. They were also asked to formulate questions in preparation for the information given by the other health professionals.

INTERVENTION WITH THE INDIVIDUAL INTERVENTION PROGRAMME

The Individual Intervention Programme for the study was the standard intervention for the target group at the low vision clinic. The participants were provided with optical aids with the aim to improve reading and near and distance viewing. Hand and stand magnifiers as well as glasses for reading were prescribed and introduced at the clinic. Assistive devices and glasses were brought home directly for practical application. Information about lighting mainly for reading was given, and the participants also received information about the disease if requested. The Individual Intervention typically included 1–2 1-h sessions at the clinic followed up by telephone contact within 2 to 4 weeks. The intervention was carried out by occupational therapists with special training in low vision.

STATISTICAL METHODS

The median, mean and range were calculated for description of the data. The 95% confidence interval (CI) for difference between participants, non-participants and dropouts were calculated for proportions using exact formulas for the binomial distribution and means[13] plus for differences between the Health Education Group and the Individual Intervention Group. The impact of the Health Education Programme on changes in perceived security in performance of daily activities was evaluated using a non-parametric statistical approach developed for evaluation of ordered categorical data.[14, 15] By this method, is it possible to measure systematic patterns of change for the group separately from the individual change unexplained by the systematic group change. In this study a measure of systematic group change, Relative Position (RP), and a measure of individual variation of change, Relative Rank Variance (RV), have been used.

Each item was evaluated in respect to the following issues:

(1) The intervention effect on each item of performance was evaluated by means of a contingency table,[14] which shows the frequency distribution of the two assessments of perceived security pre- and post-intervention. The four categories of perceived security were very insecure (1), insecure (2), quite secure (3) and secure (4). The main diagonal of unchanged level of perceived security is oriented from the lower-left to the upper-right corner in this approach, (figure 1). Figure 1a shows that 22 (19 + 3) persons out of 51 rated the same level of security/insecurity on the two occasions.

(2) The presence of systematic change in perceived security in performance is evident from the categorical distributions, also called the marginal distributions, measured on the two occasions. The marginal distribution in figure 1a is 4, 11, 15 and 21 at baseline and 0, 2, 6 and 43 at the 28-month evaluation. The difference between those marginal distributions indicate a systematic change over time, which can be visualized by plotting the cumulative proportions of those marginal distributions against each other, starting with the point (0.0). This plot forms a kind of Relative Operating Characteristic (ROC) curve.[14] A ROC curve below the diagonal indicates an improvement, (figure 1a) and a ROC curve above the diagonal indicates deterioration (figure 1c). By the ROC curve it is possible to identify and to locate the most important changes in perceived security in performance of daily activities over time. Possible values of RP range between -1 and 1, and zero value mean lack of systematic change for the group. The higher value, the higher change.[14]

(3) Dispersed observations in the contingency table, a sign of individual variations in change between the two occasions unexplained by the measures of group change, are measured by Relative Rank Variance. The measure takes into account the distances between the assessments. Figure 1b shows an example of a contingency table where the assessments are spread over the table. For example, in the activity 'writing a memo to yourself' the 7 persons assessing themselves as very insecure (1) at baseline and as secure (4) at evaluation (RV = 0.28, CI 0.09 to 0.47) are the main contributors of the individual dispersion since their rank values differ greatly. The higher the value of RV ($0 \leqslant RV \leqslant 1$), the more heterogeneous are the individual changes,[14] $RV > 0.1$ is considered cut-off value for evidence of heterogeneity. The 95% confidence intervals were calculated for the RV and RP measures using the data-based simulation bootstrap method.[16]

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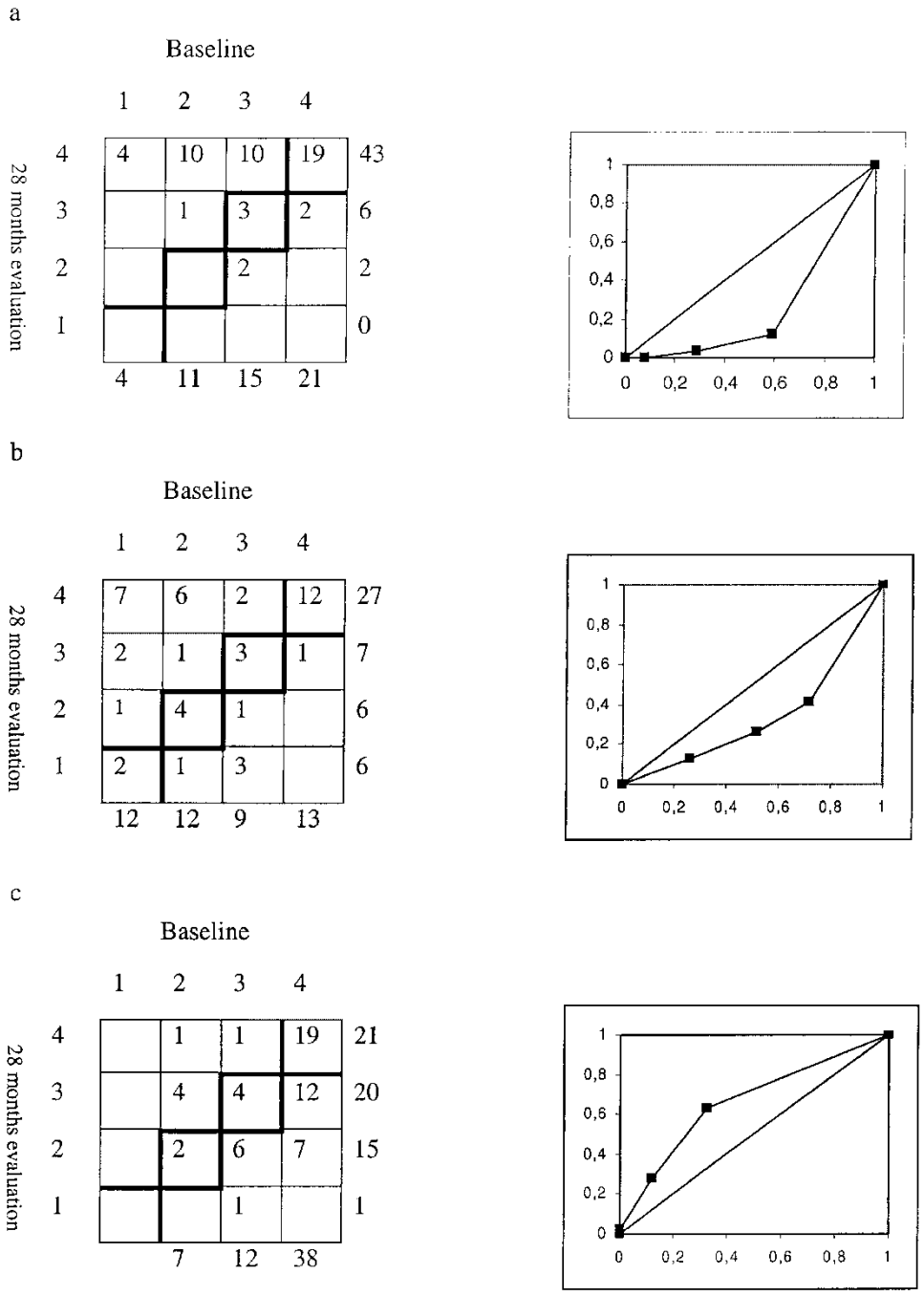


Figure 1 Contingency tables on the distribution of the frequencies of the participants' ratings of perceived security in daily occupations (1 = very insecure, 2 = secure, 3 = quite secure, 4 = secure) at baseline and at the 28-months evaluation and ROC curves of the occupations (1a) 'reading an article in your newspaper', (1b) 'writing a memo to yourself' from the Health Education Group and (1c) 'managing money and paying' from the Individual Intervention Group.

When the confidence intervals for the groups do not overlap each other, there is strong evidence for statistically significant differences between the groups. When the confidence intervals do not include the zero value,

there is strong evidence for statistically significant change between the occasions within each group.

Results

DIFFERENCES IN PERCEIVED SECURITY IN PERFORMANCE OF DAILY ACTIVITIES *BETWEEN* THE GROUPS

The study showed statistically significant differences in systematic change (RP) in perceived security between the Health Education Group and the Individual Intervention Group in several activities. The non-overlapping confidence intervals for RP between the two groups indicate statistically significant differences in perceived security in 15 activities. The values of RP varied from − 0.09 to 0.47 with a median of 0.25 in the Health Education Group. In the Individual Intervention Group the range of RP values varied from − 0.32 to 0.15 with a median of − 0.14. The Health Education Group changed towards an improved level of security compared to the Individual Intervention Group who changed towards a deteriorated level of security, as evident from the 95% CI of the RP values in table 2. For example, a systematic change towards a lower level of security in 'pouring out coffee/tea for yourself' was found in the individual intervention group (RP = − 0.04, 95% CI − 0.23 to 0.15), whereas a systematic change towards a higher level of perceived security was found in the health education group for the same activity (RP = 0.35, CI 0.21 to 0.5). The confidence intervals for the groups do not overlap each other, which is strong evidence for statistically significant differences between the groups.

Both groups showed individual changes (RV) in perceived security but no statistically significant difference was found between the groups. The values of RV among the participants in the Health Education Group varied from 0.04 to 0.32, as shown in table 2, with a median of 0.16. The RV values in the Individual Intervention Group ranged between 0.05 and 0.46 with a median of 0.1

DIFFERENCES IN PERCEIVED SECURITY IN PERFORMANCE OF DAILY ACTIVITIES *WITHIN* THE GROUPS

The Health Education Group showed statistically significant changes towards an improved level of perceived security in 20 daily activities, as both their RP values and confidence intervals were positive compared to none in the Individual Intervention Group. The Individual Intervention Group, on the other hand, showed statistically significant changes towards a lower level of security in 12 activities, as both their RP values and confidence interval are negative compared to none in the Health Education Group. For example, the Individual Intervention Group had the RP value: − 0.32 confidence interval (− 0.46 to − 0.17) which shows statistically significant change within the group in the activity 'managing money and paying' as the confidence interval does not cover the zero value. Within the Health Education Group the RP value 0.46 confidence interval (0.29 to 0.61) shows a statistically significant change (does not cover zero value) in the activity 'measuring the ingredients for making coffee' between the two occasions, baseline and 28-month evaluation, as shown in table 2.

The highest indication of group change, i.e., large RP values and a confidence interval far from zero, was found in the Health Education Group in the item 'threading a needle and sewing a button on' (RP 0.47), followed by 'measuring ingredients for making coffee' (RP 0.46) and 'reading an article in your newspaper' (RP 0.44). The systematic change in the item 'reading an article in your newspaper', as illustrated by the ROC curve (figure 1a), indicates an increased feeling of security after the intervention.

The lowest value of RP was found in the Individual Intervention Group, showing statistically significant changes towards insecurity in the item 'managing money and paying' (RP − 0.32), closely followed by 'determining if the dishes are clean' (RP − 0.28). The systematic change in 'managing money and paying' is illustrated by the ROC curve (figure 1c).

Individual variations of change unexplained by group change (RV > 0.1) were found in both groups, which means that there was evidence of heterogeneity in intervention effects in both groups. The Health Education Group showed individual variation of change in 24 daily activities, compared to 13 activities in the Individual Intervention Group. The highest values of individual variation of change in the Individual Intervention Group were found in the activities 'knowing the price of the products' (RV = 0.46), 'following the news on TV' (RV = 0.4), 'reading an article in your newspaper' (RV = 0.38) and the highest values in the Health Education Group were found in the activities 'finding your way in your local shop' (RV = 0.32), 'knowing the price of the products' (RV = 0.31) and 'reading a bank statement' (RV = 0.3) as shown in table 2.

Discussion

The purpose of the study was to investigate the impact of the Health Education Programme on perceived security in the performance of daily activity 28 months after

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Table 2 The values of Relative Position (RP) and Relative Variance (RV) in the daily activities in the Individual Intervention Group and the Health Education Group at the 28-months evaluation. Significant differences in RP *between* groups are marked in bold type. Significant differences in RP *within* groups are marked in italic

	Individual Intervention Group		*Health Education Group*		*Individual Intervention Group*		*Health Education Group*	
Daily activities	*RP*	*95% CI*	*RP*	*95% CI*	*RV*	*95% CI*	*RV*	*95% CI*
Meals								
1. pouring coffee/tea for yourself	**− 0.04**	**− 0.23 to 0.15**	***0.35***	***0.21 to 0.5***	0.35	0.19 to 0.56	0.21	0.09 to 0.37
2. finding food on the plate	− 0.11	− 0.24 to 0.01	− 0.09	− 0.24 to 0.06	0.06	0.02 to 0.13	0.13	0.05 to 0.26
3. finding things on the table while eating	− 0.09	− 0.22 to 0.03	0.08	− 0.07 to 0.22	0.06	0.02 to 0.14	0.1	0.04 to 0.22
4. finding utensils and supplies in cabinets	**− 0.08**	**− 0.23 to 0.06**	***0.22***	***0.1 to 0.35***	0.12	0.04 to 0.24	0.05	0.01 to 0.12
5. measuring ingredients for making coffee	− 0.08	− 0.23 to 0.08	*0.46*	*0.29 to 0.61*	0.11	0.04 to 0.22	0.13	0.04 to 0.3
6. determining if the vegetables are clean	***− 0.14***	***− 0.27 to − 0.02***	***0.25***	***0.06 to 0.45***	0.07	0.02 to 0.16	0.28	0.12 to 0.49
7. managing the knobs on the stove	**− 0.14**	**− 0.29 to 0.01**	***0.25***	***0.06 to 0.43***	0.13	0.04 to 0.25	0.28	0.13 to 0.48
8. slicing bread	*− 0.17*	*− 0.32 to − 0.01*	0.13	− 0.06 to 0.31	0.14	0.04 to 0.28	0.14	0.04 to 0.3
9. determining if the dishes are clean	***− 0.28***	***− 0.42 to − 0.15***	***0.18***	***0.03 to 0.34***	0.08	0.02 to 0.18	0.12	0.04 to 0.26
Self-care and care of clothing								
10. cutting/filing your nails	− 0.12	− 0.27 to 0.03	0.25	0.08 to 0.41	0.09	0.03 to 0.20	0.17	0.07 to 0.34
11. discovering if your clothes are stained	− 0.26	− 0.39 to − 0.13	0.18	0.03 to 0.34	0.09	0.02 to 0.18	0.2	0.08 to 0.36
12. threading a needle and sewing on a button	0.15	− 0.08 to 0.39	0.47	0.28 to 0.68	0.32	0.11 to 0.61	0.12	0.01 to 0.33
Communication								
13. reading an article in your newspaper	0.14	− 0.09 to 0.34	*0.44*	*0.23 to 0.63*	0.38	0.18 to 0.64	0.2	0.05 to 0.43
14. writing a memo to yourself	**− 0.06**	**− 0.24 to 0.13**	***0.33***	***0.16 to 0.5***	0.22	0.1 to 0.42	0.28	0.12 to 0.52
15. following the news on your TV	0.003	− 0.18 to 0.21	*0.34*	*0.18 to 0.5*	0.4	0.22 to 0.62	0.14	0.02 to 0.3
16. dialling on your phone	*− 0.14*	*− 0.28 to − 0.003*	0.08	− 0.06 to 0.22	0.09	0.03 to 0.18	0.11	0.04 to 0.22
Cleaning								
17. dusting your apartment	***− 0.18***	***− 0.32 to − 0.03***	***0.24***	***0.07 to 0.42***	0.07	0.01 to 0.17	0.11	0.02 to 0.25
18. vacuuming your apartment	− 0.19	− 0.38 to 0.003	0.15	− 0.09 to 0.37	0.11	0.03 to 0.27	0.14	0.03 to 0.35
Mobility								
19. going to your local shop	***− 0.17***	***− 0.31 to − 0.02***	***0.15***	***0.01 to 0.3***	0.05	0.01 to 0.12	0.04	0.01 to 0.11
20. using a pedestrian traffic light crossing	***− 0.24***	***− 0.38 to − 0.1***	***0.29***	***0.11 to 0.46***	0.09	0.03 to 0.19	0.18	0.07 to 0.35
21. distinguishing irregularity in the streets as e.g. a curb	**− 0.12**	**− 0.27 to 0.04**	***0.22***	***0.08 to 0.35***	0.18	0.07 to 0.34	0.09	0.03 to 0.21
Shopping								
22. finding your way in your local shop	*− 0.21*	*− 0.36 to − 0.06*	− 0.02	− 0.22 to 0.19	0.09	0.02 to 0.19	0.32	0.15 to 0.55
23. picking the right product	*− 0.26*	*− 0.42 to − 0.1*	0.03	− 0.14 to 0.2	0.06	0.01 to 0.16	0.15	0.05 to 0.3
24. knowing the price on the products	− 0.02	− 0.3 to 0.23	*0.35*	*0.15 to 0.54*	0.46	0.2 to 0.78	0.31	0.13 to 0.55
25. managing money and paying	*− 0.32*	*− 0.46 to − 0.17*	− 0.09	− 0.26 to 0.06	0.1	0.03 to 0.23	0.16	0.06 to 0.33
Financial management								
26. knowing your turn in the queue	***− 0.26***	***− 0.41 to − 0.13***	***0.3***	***0.12 to 0.48***	0.06	0.01 to 0.16	0.13	0.04 to 0.29
27. filling in a withdrawal form	**− 0.15**	**− 0.37 to 0.08**	***0.34***	***0.14 to 0.57***	0.24	0.08 to 0.48	0.22	0.06 to 0.47
28. reading a bank statement	0.002	− 0.17 to 0.17	*0.36*	*0.15 to 0.55*	0.09	0.02 to 0.22	0.3	0.11 to 0.57

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the intervention. There were statistically significant differences *between* the Health Education Group and Individual Intervention Group in 15 activities. *Within* the groups, the Health Education Group showed significant changes towards an improved level of security in 20 daily activities compared to none in the individual intervention group. The Individual Intervention Group showed significant changes towards a lower level of security in 13 activities compared to none in the Health Education Group. According to the findings in this study, the Health Education Programme is effective. The results are in agreement with the 4-month evaluation[4] indicating long-term stability of the improvement of perceived security in daily activities in those following the Health Education Programme.

The randomized study was set up to secure evidence-based practice in occupational therapy. According to statistical theory, randomization is a method of eliminating bias, thereby establishing that the differences between the groups are caused by the intervention. As blinding technique was not applied with regard to the intervention, group bias can occur through subconscious effects. However, this might not be a serious problem since the evaluators were not involved in the intervention and most of all because the measurement of the outcome was based on self-reports.[13] The findings in this study provide substantial support for evidence-based practice.

The non-participants and the drop-outs did not differ from the participants at baseline, apart from the non-participants who to a higher degree had been granted public transportation service. This could indicate that the non-participants were more disabled, but the other characteristics contradict this. The participants in this study seems therefore to be representative of the elderly population with age related macular degeneration that comes to the low vision clinic for their first time.

There were statistically significant differences in systematic group change (RP) *between* the Health Education Group and the Individual Intervention Group as regards 15 daily activities. Besides these differences there were significant changes of RP in improvement of 20 daily activities in the Health Education Group and deterioration in 12 daily activities in the Individual Intervention Group. This means that in five of the activities in the Health Education Group and five in the Individual Intervention Group there were additional significant group changes (RP) within the groups, apart from significant differences between the groups (those marked in italic but not bold in table 2). The Health Education Group changed towards an improvement in these activities and the Individual Intervention Group towards deterioration between baseline and the 28-month follow-up, although no statistically significant differences were found *between* the groups. There were also individual variations of change within both groups, at a level indicating individual heterogeneous changes apart from the group change. One explanation could be that the elderly population in itself is heterogeneous,[17, 18] which in this study is reflected by the wide range of age and visual acuity among the participants. Although there is no statistically significant difference between the groups in sign of individual variation of change, the individual heterogeneous changes are evidence of effectiveness on an individual level, which is of great value for practice.

The findings in this study are in agreement with the 4-month evaluation[4] demonstrating long-term stability of the improvement of perceived security in daily occupation in those following the Health Education Programme. The powerful components of the model in the Health Education Programme 'Discovering new ways' such as theoretical basis and tools for learning have been discussed more in depth in previous articles.[4, 11] However, we want to highlight the importance of client-centred practice.[2] Focus groups methodology is a method that enables the target group to participate in the planning and development of the programme and is therefore more likely to meet their needs. The advantages of the focus group methodology are that the profession can become familiar with its market and acquire knowledge of the target population and the factors that might influence daily activities. Focus group methodology gave an opportunity for collaboration and partnership[1, 19] with the target group to develop the programme. A client-centred approach involving the client in all phases of the intervention is supported in the literature[19] and might be one of the explanations why the programme has been found to be so successful so far.

The programme was developed from a broad approach combining qualitative and quantitative methods with their different advantages and strengths.[20, 21] The methods were used in sequence so that what was learned from the studies in the pre-intervention stage was added to what was learned from the studies in the intervention stage, and so on. The strengths of different methods can be successfully combined to address the complexity of the research question and when the goals of the project include both pure research and applied uses in practice.[21] Further, evidence-based practice uses research evidence to make decisions about intervention that are effective for the specific target group.[2] The metho-

dology used here can be highly recommended for developing programmes within practice.

The long-term evaluation provides evidence that the Health Education Programme supports the elderly person by enabling change. Studies show[17, 18] that elderly persons with visual impairment have multiple health problems and that all those health problems must be addressed when developing a programme. The Health Education Programme described in this article is one example of a programme tailored to meet the needs of elderly with AMD based on the target group's perceived security in daily activities. When developing a programme for elderly persons, it is vital to focus not only on one impairment but to see the person as an integrated whole, considering the multiple health problems of the population and how they affect daily activities. The study has shown that an intervention such as the one described above is effective, allowing people to continue to live with a chronic progressive disease independently. Further research will be conducted in order to investigate whether the programme has any impact on perceived health and morbidity. A health economic evaluation will also be made of the programme.

In conclusion, the findings provide strong support for the long-term effect of the programme for persons with AMD. There is strong evidence that the Health Education Programme enhances security and hinders a progressive decline in perceived security in daily activities, thereby providing the material necessary for evidence-based practice.

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